RDM

RDM-7 DASH QUICK START GUIDE

From the box to the car: first boot, CAN setup, channels, units and the web editor.

GET UP & RUNNING · FIRST BOOT → WIZARD → CONNECT → BUILD → DRIVE

DASH HOTSPOT (WIFI NAME)

RDM7-XXXX

HOTSPOT PASSWORD

Unique per unit — shown in the setup wizard & WiFi Settings

EDITOR ADDRESS (HOTSPOT)

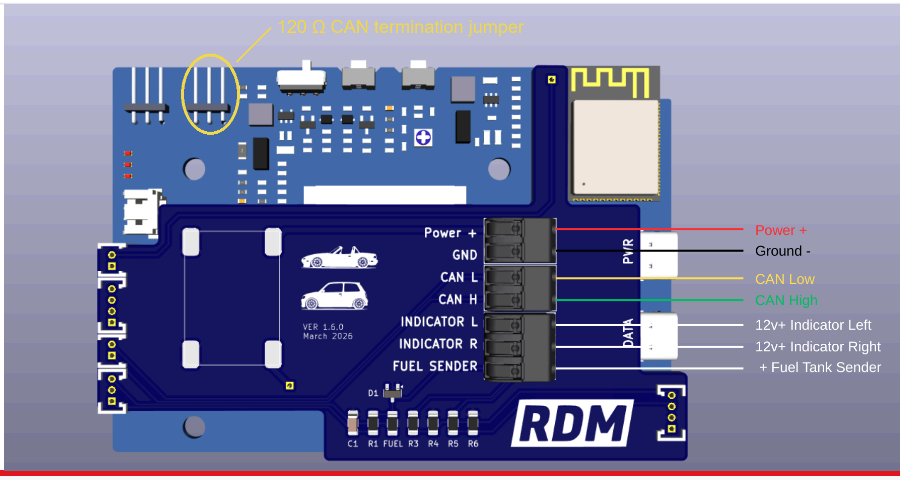
`http://192.168.4.1`

EDITOR ADDRESS (HOME WIFI)

Shown in Device Settings

01 WIRING THE DASH

Everything connects to the quick-connect push terminals on the back of the dash — strip the wire, push it in, done. Get these few wires right and the rest of this guide is pure software.



WIRING MAP The back of the board — quick-connect push terminals on the right, 120 Ω CAN termination jumper circled top-left.

TERMINAL	CONNECT TO
Power + / GND	Switched 12 V (ignition-on) feed and chassis ground.
CAN H / CAN L	Your ECU or vehicle CAN bus — use a twisted pair.
INDICATOR L / R	12 V+ inputs — tap into your left and right indicator circuits to drive the on-screen turn arrows.
FUEL SENDER	The sense wire straight from your fuel tank float — the resistor is built into the dash, nothing to add. Sender ground goes to a sensor ground or ties in with the dash's ground. Calibration is in Section 09.

CAN TERMINATION

A CAN bus needs a **120 Ω** terminating resistor at each end. If the dash sits at the end of the bus, enable its built-in one with the jumper at the top-left of the board (circled in the photo — it's the right-hand one of the two, marked with a yellow tab on the back).

02 FIRST BOOT & THE SETUP WIZARD

With the dash wired up (Section 01), key on. The very first time it boots, the RDM-7 runs its on-screen Setup Wizard and does most of the work for you.

1 **CAN scan**

The dash listens on the bus and automatically detects the bitrate (125k / 250k / 500k / 1M). Make sure the engine is on, or at least the ignition, so the bus is talking. If the scan comes up empty, work through the CAN checklist in Troubleshooting at the back of this guide (Section 12).

2 **ECU auto-detect**

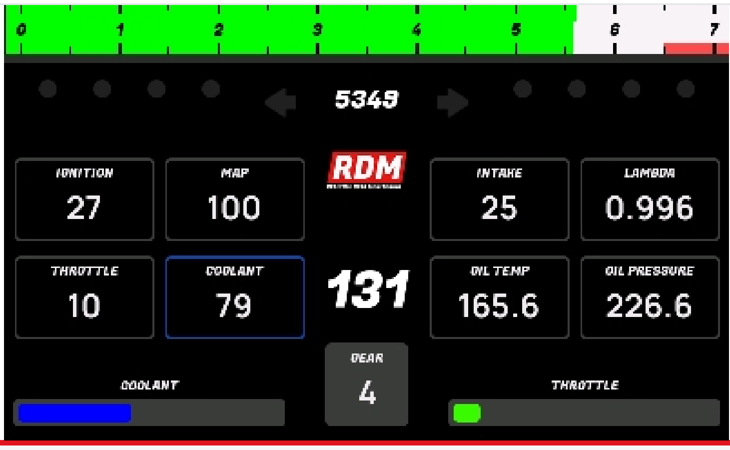
The CAN traffic is matched against the built-in ECU library and the best match is shown with a confidence score. If it guessed wrong, tap **Pick a different ECU** and choose from the list.

3 **Channels review**

The wizard binds every signal the ECU broadcasts (RPM, coolant, MAP, …) to named channels. Skim the list — tap any channel to point it somewhere else.

4 **WiFi**

Pick **Join WiFi Network** or **Start Hotspot** — either one finishes the wizard and drops you on the dashboard. Section 03 covers what each means.



ON THE DASH Up and running — the default dashboard driving live CAN data.

BUILT-IN ECU PRESETS

PRESET	GOOD FOR
ECU Master Black / Classic	EMU Black & Classic standard broadcast
MegaSquirt MS3-Pro	MS3-Pro standard CAN broadcast
Haltech Nexus / Elite	Haltech V2 protocol

PRESET	GOOD FOR
MaxxECU 1.2 & 1.3+	MaxxECU dash broadcast (1.3 adds oil/fuel pressure)
Link ECU Generic Dash	Link generic dash stream
Ford Falcon BA/BF & FG	Factory Falcon CAN buses
Toyota 86 / Subaru BRZ	Factory HS-CAN, 2012–2020
OBD2 Standard	Any 2008+ car via the OBD2 port — no aftermarket ECU needed

> ***NOTE***
>
> You can re-run the wizard any time from **Setup → Re-run Setup Wizard**. Changing ECU preset later automatically re-points your existing widgets at the new ECU's signals.

03 GET CONNECTED

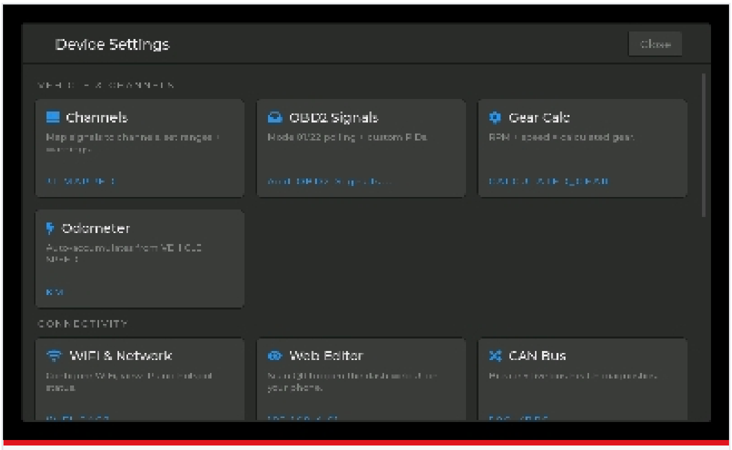
The dash talks to your phone or laptop over WiFi in one of two modes — its own hotspot, or joined to your home network. It's one or the other: pick it from the **Mode** dropdown in **Setup → Device Settings → WiFi & Network**, and switch whenever you like.

▸ HOTSPOT MODE (ZERO SETUP — IN THE CAR)

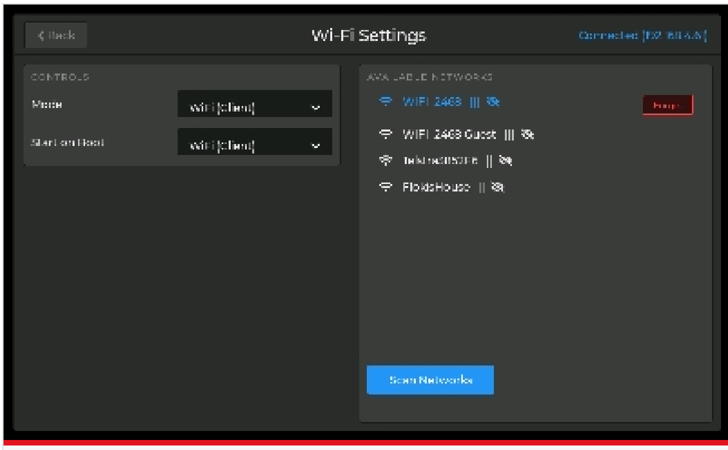
Pick **Start Hotspot** in the wizard, or set **Mode** to Hotspot in WiFi Settings. Then join the WiFi network named **RDM7-XXXX** (the XXXX is unique to your unit) — the password is unique to your unit and shown right on the dash (setup wizard, and any time under WiFi Settings); you can change it there too. A captive-portal prompt usually pops up and lands you straight in the editor; if it doesn't, browse to `http://192.168.4.1`.

▸ CLIENT MODE (JOIN YOUR HOME NETWORK)

Pick **Join WiFi Network** in the wizard, or set **Mode** to **WiFi (Client)** in WiFi Settings: choose your network from the scan list and type the password. Once connected, the dash's address appears at the top of that screen — browse to it from anything on the same network, and bookmark it. There's a **QR code** too if your phone is closer than your keyboard. (Can't reach the dash? Your router's client list will show it as **RDM7-...**.)



STEP 1 Device Settings — tap the *WiFi & Network* card.

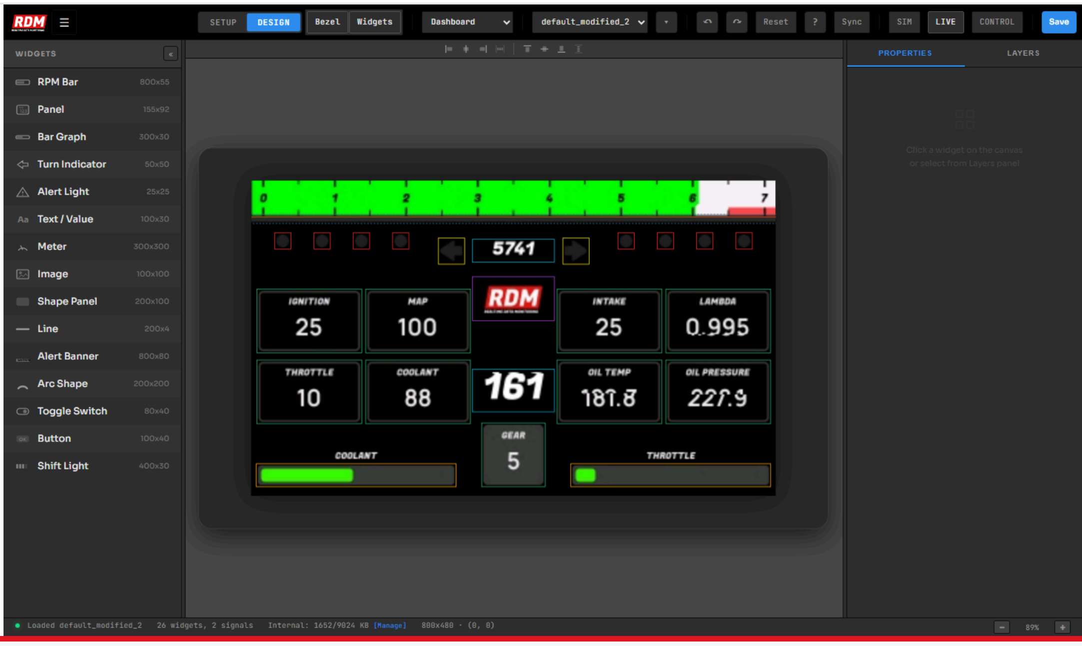


STEP 2 Pick a network from the scan list — your dash's address shows top-right once connected.

04 THE WEB EDITOR AT A GLANCE

Browse to the dash's address and the **RDM-7 Visual Designer** loads — this is where you build screens and configure data. Changes appear on the real dash the moment you hit save.

- **Widgets palette (left)** — drag gauges, bars, panels, text, indicators, warning lights, images and more onto the canvas.
- **Canvas (center)** — your dashboard, live. Drag to position, use the alignment tools to line things up. The canvas previews **real values from the car** by default (toggle with the LIVE pill).
- **Inspector (right)** — select any widget to edit its channel, range, colors, fonts and alert behavior.
- **SAVE** — writes the layout to the dash and reloads the screen instantly.
- **Menu (☰)** — Channels, Live Signals, Custom CAN Signals, OBD2 Setup, ECU Preset, Data Logger, Gear Setup, WiFi, Device Info and Diagnostics.



WEB EDITOR The Visual Designer — widget palette left, live canvas center, Inspector right, SAVE top-right.

> ***TIP***
>
> Hit **SAVE** and the dash reloads its screen on the spot — so you can tweak a layout with the car running and watch it change in real time.

05 CHANNELS — WIRING DATA TO WIDGETS

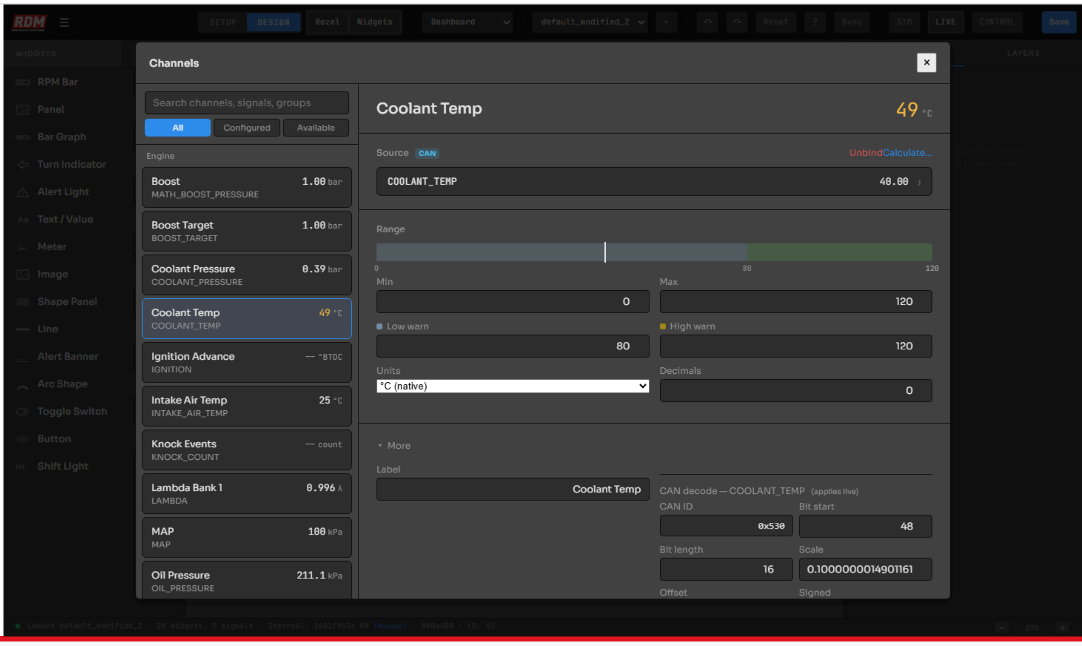
A **channel** is a named value like *RPM*, *Coolant Temp* or *Boost*. The channel owns where the data comes from (ECU CAN, OBD2, custom CAN, or the dash's own inputs) plus its units, range and warning thresholds. Widgets just point at a channel — so you can swap data sources without touching your layout.

▸ ASSIGN A CHANNEL TO A WIDGET (WEB)

1. Click the widget on the canvas — the Inspector opens on the right.
2. Click the **Channel** field — the channel picker opens. Search or scroll, then select.
3. The widget immediately follows that channel's live value, units, decimals and thresholds.

▸ CHANGE WHERE A CHANNEL GETS ITS DATA

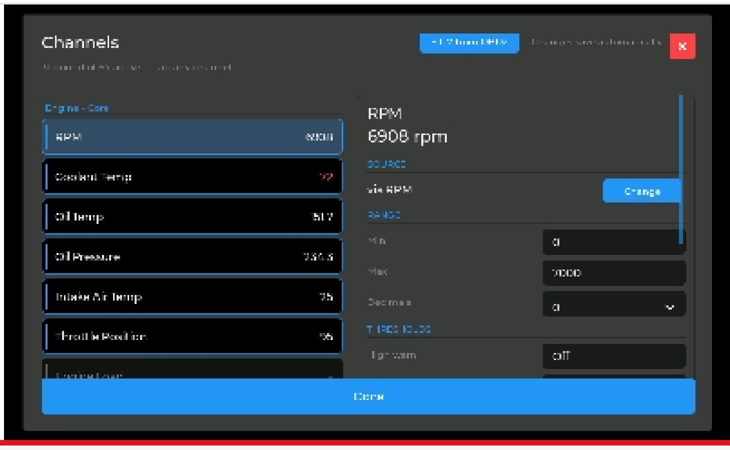
1. Open `☰ → Channels`. Use the **All / Configured / Available** filter to find what you need.
2. Select a channel — its detail pane shows the current source as a badge (**CAN**, **OBD2** or **RDM-7**).
3. Hit **Change** on the source to pick a different ECU preset signal, an OBD2 PID, or a fully custom CAN decode.



WEB EDITOR The Channels modal — source badge and live value up top, Range with warn thresholds, Units & Decimals, and the raw CAN decode under *More*.

ON THE TOUCHSCREEN

The same channel editor lives on the device: **Setup → Channels**. Tap a channel to edit its source, range, units and warning thresholds with the on-screen keypad — no laptop needed.



ON THE DASH The on-device channel editor — list on the left, live value, source, range and thresholds on the right.

TIP — CALCULATED CHANNELS

In the channel detail you'll also find **Calculate**: build a channel from two others (or a channel and a number) with + − × ÷. Classic example: *Boost* = *MAP* − *Baro*. Units are handled for you.

06 CHANGING UNITS & DECIMALS

Prefer °F over °C, psi over kPa, mph over km/h? Units live on the channel, so changing them once updates every widget bound to it.

IN THE WEB EDITOR

1. Open `☰ → Channels` and select the channel (e.g. *Coolant Temp*).
2. In the **Range** section, pick the display unit from the **Units** dropdown and set **Decimals**.
3. Min / Max and warning thresholds in this section are shown and edited **in your chosen unit** — type what you see on the gauge.

ON THE TOUCHSCREEN

1. Go to `Setup → Channels` and tap the channel.
2. In the **Range** section, pick the unit from the dropdown — the whole page recalculates instantly, including the live value.
3. Edit range and thresholds with the keypad, in the displayed unit.

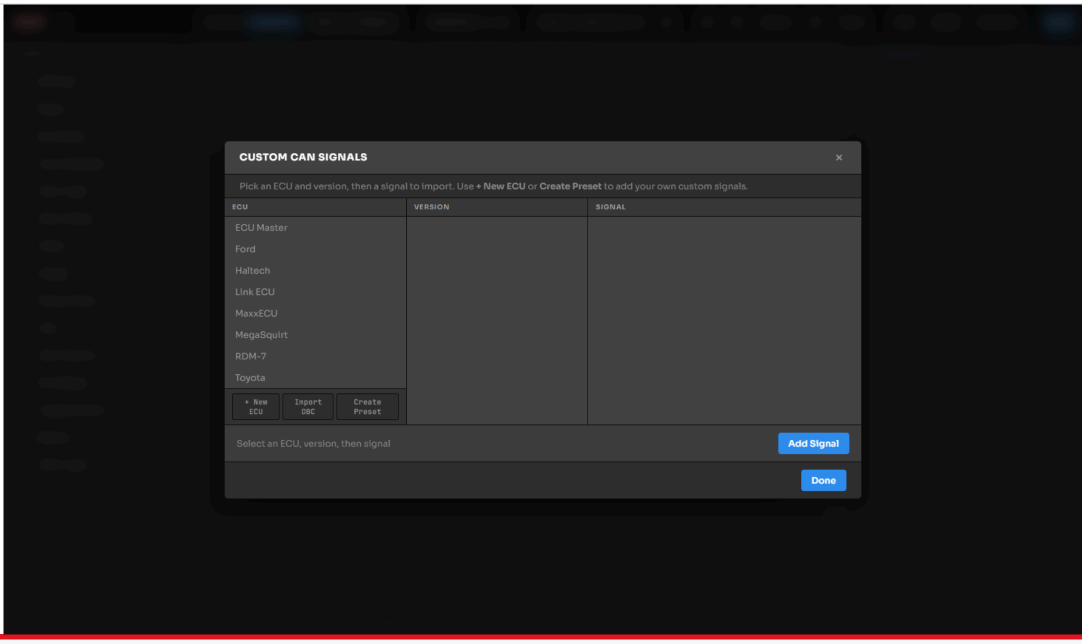
> ***NOTE***
> Conversion is automatic and live — the dash keeps the raw sensor data untouched and converts for display, so switching back and forth never degrades your numbers.

07 CUSTOM CAN SIGNALS

Got a sensor module, a non-listed ECU, or a one-off CAN message? Decode anything on the bus yourself.

ADD A ONE-OFF SIGNAL

1. Open `☰ → Custom CAN Signals`. The browser shows the preset catalog; tap **+ Add CAN Signal** to create your own.
2. Fill in the decode: **CAN ID** (hex), **bit start**, **bit length**, **scale**, **offset**, byte order (**Intel/LE** or **Motorola/BE**) and whether it's **signed**.
3. Save, then bind a channel to it (Section 05) and drop a widget on it. The value goes live immediately — watch it in `☰ → Live Signals` while you dial the decode in.



WEB EDITOR Custom CAN Signals — browse the ECU catalog, or use + *New ECU*, *Import DBC* and *Create Preset* to roll your own.

IMPORT A DBC FILE

If you have a `.dbc` database for your vehicle or device, import it from the menu — every message and signal inside becomes available to bind, no manual bit-math required.

SAVE YOUR OWN PRESET

Built a full set of decodes for an unlisted ECU? Save it as a **custom preset** — it appears alongside the factory presets so you (or your next car) can apply the whole set in one tap.

08 OBD2 — FILLING THE GAPS

On a factory car (2008+), the dash can poll standard OBD2 PIDs over the same CAN connection — either as the main data source or to back-fill what your ECU doesn't broadcast.

- **Whole-car OBD2:** pick the **OBD2 Standard** preset in the wizard — ~30 common PIDs are polled automatically.
- **Fill from OBD2:** in a channel's detail pane, tap the **Fill from OBD2** chip — the dash probes the car for a matching PID and binds it if found. Great for adding intake temp or fuel level next to ECU data. (No fuel PID on your car? Wire a sender directly — see Section 09.)
- **OBD2 Setup** (**☰ → OBD2 Setup**) lets you browse supported PIDs and manage polling; **Diagnostics** includes a DTC fault-code reader.

> ***NOTE***
>
> Channels show a badge for their source — **CAN**, **OBD2** or **RDM-7** (internal, e.g. calculated gear) — so you always know where a number comes from.

09 FUEL LEVEL SENDER

Most aftermarket ECUs don't broadcast fuel level, and not every car exposes it over OBD2. The dash reads a resistive tank sender directly — wire two wires, then teach it your tank with a quick calibration.

CONNECT THE SENDER

- Run the sense wire from your fuel tank float straight into the **FUEL SENDER** terminal (see the wiring map in Section 01). The resistor is built into the dash — no external components needed.
- The sender's ground goes to a sensor ground, or ties in with the same ground as the dash.

CALIBRATE IT

You don't enter resistances — you capture the live voltage at tank levels you already know. Open **☰ → Fuel Sender** in the web editor (or the **Fuel Sender** card in Device Settings on the dash):

1 Set your end points

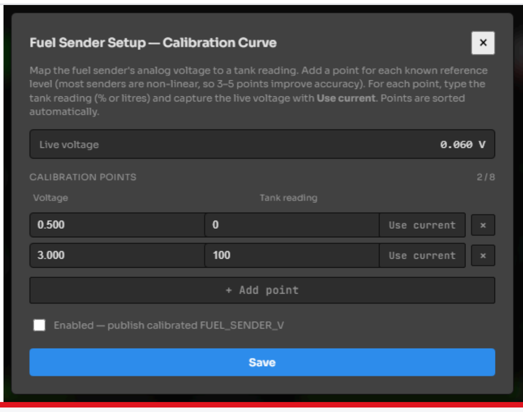
With the tank empty, set **Tank reading** to `0` and tap **Use current** to grab the live voltage. Fill the tank, set the next point to `100` (percent) or your tank's litres, and tap **Use current** again.

2 Add mid points (optional)

Senders are rarely linear. For a truer gauge, add points in between — quarter, half, three-quarter — each captured with **Use current**. Up to 8 points; the dash sorts them and interpolates between.

3 Enable & save

Tick **Enabled — publish calibrated FUEL_SENDER_V** and hit **Save**. Leave it unticked to read raw voltage while you're testing.



WEB EDITOR The calibration curve — capture live voltage at known tank levels; add up to 8 points for non-linear senders.

> **NOTE**
>
> Once enabled, the calibrated level is just another signal — bind a bar, gauge or panel to your **Fuel Level** channel, set its units (% or litres) and add a low-fuel warning like any other channel.

10 ALERTS & THRESHOLDS

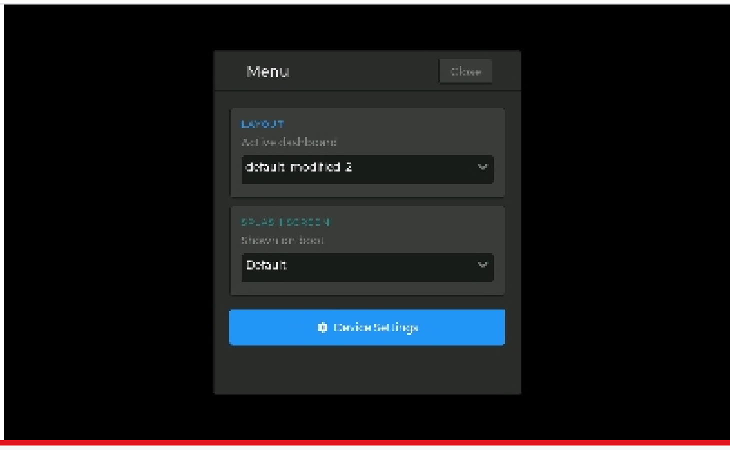
Warning limits also live on the channel — set them once, and every gauge, bar and panel bound to that channel reacts.

1. Open the channel (web **☰ → Channels**, or **Setup → Channels** on-device).
2. Set **Warn Low** and/or **Warn High** in the Range section — in your display units.
3. On the dash: arcs and meters show a redline zone, bars and panels change color, and **warning-light widgets** trip when the value crosses the limit.

> ***TIP***
>
> Each widget controls its own alert *colors* in the Inspector — the channel decides *when*, the widget decides *how it looks*.

11 EVERYDAY FEATURES

FEATURE	WHERE	WHAT IT DOES
Data logger	**☰ → Data Logger**	Records all channels to CSV. Set a rate (or Max), start/stop, and download logs straight from the browser. **Requires an SD card** fitted in the dash.
Gear indicator	**☰ → Gear Setup**	Calculates current gear from RPM + speed. Enter tire size, final drive and gear ratios once.
Layouts	Editor toolbar	Save multiple dashboards (street / track / debug) and switch between them. Layouts are portable between RDM-7 units — your car-specific CAN setup stays on the device.
Screen rotation	**Setup → Device Settings**	Rotate the display for your mounting orientation.
Firmware updates	Editor banner	When an update is available the editor shows a banner at the top — one click installs it over WiFi, with a live progress bar. The dash keeps the previous version as a fallback if anything goes wrong. (Current version is under **☰ → Device Info**.)



ON THE DASH Tap the screen, then the *Menu* pill — switch dashboards, set the boot splash, or dive into Device Settings.

12 TROUBLESHOOTING

NO CAN DATA? WORK DOWN THIS CHECKLIST

1 Wake the bus

Ignition on — better yet, engine running. Most buses are silent with the key out.

2 Check the pair

CAN H and CAN L are easy to get backwards — swapping them won't damage anything, so if you're seeing nothing, swap and retry.

3 Termination

A CAN bus needs a **120 Ω** resistor at each end. If the dash is at the end of the bus (e.g. wired straight to an ECU), enable its built-in termination with the jumper at the top-left of the board — the right-hand one of the two 3-pin headers, marked with a yellow tab on the back (circled in the Section 01 photo).

4 Bitrate

Re-run the bitrate scan from **Setup → Device Settings** — a wrong bitrate looks exactly like a dead bus.

5 Wiring quality

Use a twisted pair for CAN H/L, keep it away from ignition leads, and make sure grounds are solid.

SYMPTOM	TRY THIS
Gauges show **--**	The channel has no fresh data. Check ignition/engine is on, then **☰ → Live Signals** to see what's arriving. Values go stale after 2 seconds of silence.
Wizard found no CAN traffic	Work down the CAN checklist above — wake the bus, swap H/L, check 120 Ω termination, re-scan the bitrate.
Wrong / weird values	Usually a decode mismatch — confirm the ECU preset matches your ECU firmware's broadcast version, or check scale/offset/endian on a custom signal.
Can't reach the editor	Confirm you're on the same network as the dash. Fall back to the hotspot: join **RDM7-XXXX** and open `192.168.4.1`.
Dash dropped off home WiFi	The dash keeps driving the gauges and falls back to its own hotspot — join it, open WiFi Settings in the editor and re-enter the password.
Fuel gauge pinned or wrong	Open **☰ → Fuel Sender** and watch **Live voltage**. If it doesn't move with fuel level, check the sense wire and the sender's ground. Otherwise recapture your empty and full points.
Widget ignores a units change	Make sure the widget is bound to the channel (Inspector → Channel) rather than carrying its own old settings — rebind and save.

QUESTIONS? BUILDS?
GET IN TOUCH.

SUPPORT & MORE INFORMATION

realtimedatamonitoring.com.au

QUESTIONS & QUERIES

info@realtimedatamonitoring.com.au

SHOW US YOUR BUILD

Tag @realtimedatamonitoring on Instagram!

Scan the code for the latest firmware, layouts and documentation. And when your dash is in the car and the gauges are alive — tag us. Seeing your builds is the best part of this job.